BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 163 · Final Targeted Revision

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is a logic error?

Q2. How many different codes can 7 bits represent?
A. 7
B. 64
C. 128
D. 256

Q3. How does a bitmap represent pixel colours?

Q4. An image file is larger than its raw pixel data. “Metadata is just the colour of each pixel.” Correct this. How can metadata affect file size?

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I Do · Read new knowledge

Key vocabulary

Personalised revision — Choose priorities from a specific error log, not assumptions.

Interleave topics — Alternate a calculation, written explanation and algorithm trace.

Self-assessment — Mark answers against explicit expected points.

Retest — Move to new questions after correction so improvement transfers.

New learning

Personalised revision: Choose priorities from a specific error log, not assumptions.

Interleave topics: Alternate a calculation, written explanation and algorithm trace.

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Worked example

A pupil missed both virtual memory and binary-search prerequisite. Write two improvements.

1. Virtual memory: OS stores some pages on secondary storage when RAM insufficient.

2. Secondary storage accesses are slower than RAM, reducing performance if paging is frequent.

3. Binary search: array must be sorted.

4. Midpoint comparisons then safely eliminate half the range.

We Do · Practise with guidance

Guided 1

Correct a mixed two-mark answer: 'ROM is volatile and holds open apps'.

Worked method: Correct both terms accurately.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Complete the OCR ERL conversion that displays the character for ASCII code 65.

```
code = 65
print(...)
```

write

Question 1

Explain the difference between RAM and virtual memory. [4 marks]

Question 2

What is 11110000 in denary? [2 marks]

Question 3

State why a merge step requires sorted sublists. [3 marks]

Exit Ticket · Answer before checking

Repeating a copied answer is stronger evidence than independently solving a new question.

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Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. A mistake in the algorithm that gives the wrong result, even though the code follows the language's rules.

Q2. C — 128

Q3. Each pixel has a colour code stored in binary.

Q4. Metadata is extra information, such as width, height or colour depth. It adds to raw pixel data, so the real file may be larger.

I Do / We Do pages · Guided practice

Q1. Correct both terms accurately.

You Do pages · Code First

```
code = 65
print(CHR(code))
```

Expected result: A

You Do pages · Independent questions

Q1. RAM is physical primary working memory; virtual memory uses secondary storage to back pages when RAM is insufficient.

Q2. 240.

Q3. Choosing the smaller front item produces an ordered result only when each incoming sublist is already ordered.

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question: